Restricted *abc* hypotheses on the scalar families

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Abstract

We examine two unconditional restricted estimates on scalar families formed from elliptic traces. For the Frey curves of a sum of even powers, the reduced denominator of the j-invariant is exponentially large in its logarithmic height. Consequently a polynomial bound for that denominator cannot hold uniformly in the scalar family. We also show that the fixed asymmetric-size hypotheses of two subexponential *abc* estimates fail along infinitely many required integer witnesses. The conclusions identify the hypotheses that fail on these scalar families and give exact quantitative bounds for that failure.

1 The denominator of the Frey invariant

For $r \in \mathbb{Q}$, let $h(r)$ be its logarithmic projective height and $\mathrm{den}(r)$ its positive reduced denominator. Fix a positive even integer d. For nonzero coprime integers A, B, put

$$a = A^d > 0, \qquad b = B^d > 0, \qquad c = a + b, \qquad E_{A,B,d} : y^2 = x(x-a)(x+b).$$

The curve is nonsingular since $abc \neq 0$.

Proposition 1.1. *Let $H = \max(|A|, |B|)$. Then*

$$j(E_{A,B,d}) = \frac{256(a^2+ab+b^2)^3}{a^2b^2c^2}.$$

The common divisor of this displayed numerator and denominator is

$$2^{\min(8, v_2(a^2b^2c^2))}.$$

In particular,

$$\mathrm{den}(j) \geq \frac{H^{4d}}{256}, \qquad h(j) \leq 6d \log H + \log 6912. \tag{1}$$

Consequently

$$\log \mathrm{den}(j) \geq \frac{2}{3} h(j) - \frac{2}{3} \log 6912 - \log 256. \tag{2}$$

Proof. The Weierstrass invariants are

$$c_4 = 16(a^2+ab+b^2), \qquad \Delta = 16a^2b^2(a+b)^2,$$

which give the formula. The positive integers a, b, c are pairwise coprime. If an odd prime divides a, the expression a^2+ab+b^2 reduces to b^2 and is a unit; the analogous statement holds for b. If it divides c, then $a \equiv -b$ and the same expression again reduces to b^2. Thus no odd prime cancels. The expression is odd: if one of a, b is even, the other is odd, and if both are odd its three summands are odd. Hence the exact cancellation is the displayed power of two.

Since $|AB| \geq H$ and $c \geq H^d$, the original denominator is at least H^{4d}, proving the first bound. The numerator is at most $256(3H^{2d})^3 = 6912H^{6d}$, and the denominator is at most $H^{4d}(2H^d)^2 = 4H^{6d}$. Reduction cannot increase either number, which proves the second bound. Substitute the resulting lower bound for $\log H$ into the first bound to obtain (2). □

Corollary 1.2. *For fixed constants $C > 0$ and $k \geq 0$, only finitely many primitive pairs (A, B) satisfy*

$$\operatorname{den}(j(E_{A,B,d})) \leq C h(j(E_{A,B,d}))^k.$$

Proof. The first bound in (1) shows that $h(j)$ tends to infinity with H. The exponential lower bound (2) eventually exceeds every fixed polynomial in $h(j)$. Thus H is bounded. □

Pasten's restricted Szpiro theorem [1, Theorem 1.2] assumes a bound for $\operatorname{den}(j)$ by a fixed power of $\log|\operatorname{num}(j)|$. Replacing this logarithm by $h(j)$ weakens the hypothesis. Corollary 1.2 therefore excludes both hypotheses on an unbounded family of these Frey curves. Passing to another Weierstrass model or taking a quadratic twist leaves j unchanged and does not repair this problem.

2 Application to the elliptic scalar

Put $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, and define

$$E : y^2 = x^3 - (3-\rho)^2 x, \qquad P = 2(1+\rho, 2\rho) = \left(\frac{57-28\rho}{8}, \frac{-660+417\rho}{32}\right).$$

The point P has infinite order [4, Lemma 8]. Put

$$T(n) = \operatorname{Tr}_{K/\mathbb{Q}} x([n]P), \qquad n \neq 0.$$

Theorem 10 of [4], with one point, gives constants $A_0 > 0, J_0 \geq 0$ such that

$$n^2 \leq A_0 h(T(n)) + J_0. \tag{3}$$

Indeed $\widehat{h}_E([n]P) = n^2 \widehat{h}_E(P)$ and changing the half-trace to the trace changes logarithmic height by at most $\log 2$. Also $T(n) > 6$: every multiple of P is a double, its abscissa is a square in K by the duplication formula, and the curve equation gives $x([n]P) > 3 \mp \sqrt{2}$ at the two real embeddings. Consider the scalar pair $(n, T(n))$. Write $n/T(n) = A_n/B_n$ in lowest terms. Then

$$\log \max(|A_n|, |B_n|) = h(n/T(n)) \geq h(T(n)) - \log|n| \geq \frac{n^2 - J_0}{A_0} - \log|n|.$$

This tends to infinity. Corollary 1.2 consequently applies to these scalar pairs for every unbounded integer polynomial sequence of indices. This includes the families used to represent integer inputs by shifts of squares or shifts of fourth powers.

3 The asymmetric-size restrictions

Theorem 1.3 of Pasten–Sepúlveda-Manzo [2] applies to primitive positive triples $a + b = c$ satisfying, for a fixed $\tau > 0$,

$$\min(a, b) \leq c \exp\bigl(-(\log c)^{\tau} \log_2^{*} c\bigr). \tag{4}$$

The earlier restricted estimate recalled in their Theorem 1.2 instead uses $\min(a, b) \leq c^{1-\eta}$ for a fixed $\eta > 0$. The word "fixed" is essential for a uniform height conclusion.

Lemma 3.1. *Let $p \in \mathbb{Z}[X]$ be nonconstant, tend to positive infinity, and satisfy $p(0) \neq 0$. There are arbitrarily large integers m for which*

$$6 < T(p(m)) \leq C_p$$

with a constant C_p independent of m.

Proof. Consider the point $(P, \sigma P)$ in the compact real Lie group $E(\mathbb{R}) \times E^{\sigma}(\mathbb{R})$, and the closure G of its cyclic subgroup. The identity component G^0 is a torus, and G/G^0 is finite. Choose a positive integer q killing this component group. Continuity and compactness show that the closure of the group generated by $q(P, \sigma P)$ is qG. This equals G^0: it is contained in G^0, while multiplication by q is surjective on the torus G^0. For $m = qk$, the quotient $r(k) = (p(qk) - p(0))/q$ is a nonconstant integer polynomial. For each nontrivial character of G^0, its value on $q(P, \sigma P)$ is an irrational rotation, by density and connectedness. Weyl's polynomial equidistribution theorem [3], applied to each character and the polynomial r, therefore shows that $[p(qk)](P, \sigma P)$ is equidistributed in the coset $[p(0)](P, \sigma P) + G^0$.

The point $[p(0)]P$ and its conjugate are finite nontorsion points. Choose a relatively open neighborhood of this tuple whose compact closure avoids the two identity divisors. The trace of the abscissas is bounded above on that closure. The neighborhood has positive Haar measure in the coset and is visited infinitely often. Positivity gives the lower bound six at all the resulting nonzero multiples. □

Proposition 3.2. *Fix a positive even d and a polynomial input sequence as in Lemma 3.1. Along infinitely many indices $n = p(m)$, the primitive triple*

$$a = A_n^d, \qquad b = B_n^d, \qquad c = A_n^d + B_n^d, \qquad A_n/B_n = n/T(n),$$

fails (4) for every fixed $\tau > 0$, eventually along that subsequence. It also fails $\min(a, b) \le c^{1-\eta}$ for every fixed $\eta > 0$.

Proof. Take the subsequence where $6 < T(n) \le C_p$. For sufficiently large n, $A_n > B_n > 0$, and

$$\log \frac{c}{\min(a, b)} = \log \left(1 + \left(\frac{n}{T(n)} \right)^d \right) = O(\log n).$$

By (3) and the preceding lower bound for $h(n/T(n))$,

$$\log c \ge d \log A_n \ge d \left(\frac{n^2 - J_0}{A_0} - \log n \right) \gg n^2.$$

For every fixed $\tau > 0$, the quantity $(\log c)^{\tau} \log_2^* c$ grows faster than $\log n$. This contradicts the logarithmic form of (4). Likewise $\eta \log c \gg n^2$ exceeds the displayed logarithmic ratio, proving the second assertion. □

Thus the cited restricted estimates cannot be invoked uniformly on these scalar triples. A uniform application would require a different scalar family or a restricted estimate with different hypotheses. Failure of the size hypotheses says nothing about the validity of the ordinary *abc* inequality on these triples.

4 Exact verification

The companion script `check_frey_denominator.py` computes the Weierstrass invariants independently, verifies the rational formula and its exact two-adic cancellation on a finite exhaustive set, and checks the integer forms of the two displayed height majorants. The arguments proving all-input statements and infinite subsequences are the proofs above; finite enumeration is not substituted for them.

References

[1] H. Pasten, *Szpiro's conjecture when the denominator of the j-invariant is small*, CUBO **28** (2026), no. 2, 383–389, https://doi.org/10.56754/0719-0646.2802.383. Primary manuscript: https://arxiv.org/abs/2308.07114v2, Theorem 1.2 and Corollary 3.4.

[2] H. Pasten and R. Sepúlveda-Manzo, *On the abc and the abcd conjectures*, Bull. Braz. Math. Soc. (N.S.) **56** (2025), article 54, https://doi.org/10.1007/s00574-025-00476-1. Primary manuscript: https://arxiv.org/abs/2406.05083v1, Theorems 1.2 and 1.3. Only their precise size hypotheses are used here.

[3] H. Weyl, *"Uber die Gleichverteilung von Zahlen mod. Eins*, Math. Ann. **77** (1916), 313–352, https://doi.org/10.1007/BF01475864.

[4] Project repository, *Aggregate trace heights of elliptic multiples*, 8 November 2026, Lemma 8 and Theorem 10, `papers/trace_heights/aggregate_trace_height_20261108_v9.tex`. The height comparison applies J. Von Buhren, *Borne de hauteur semi-effective pour le problème de Mordell–Lang dans une variété abélienne*, J. Théor. Nombres Bordeaux **29** (2017), 307–320, Theorem 1.1, https://doi.org/10.5802/jtnb.981.